NEXUS ERP

Supervisor Meeting Brief

Student: Nethmina Kavinda Bandara

Programme: BSc (Hons) Computer Science

University: NSBM Green University

Methodology: Design Science Research (Peffers' DSRM)

Date: 2 April 2026

Project Title: Design and Development of an Integrated ERP Portal with Employee Risk Scoring Using Synthetic Social Behaviour Signals

Tech Stack: MERN (MongoDB, Express.js, React.js, Node.js) + Python Flask ML Microservice

1. Project Overview

NEXUS ERP is a web-based Enterprise Resource Planning system for SMEs that integrates four core modules into a single platform:

- HR / Employee Management - employee profiles, roles, CRUD operations
- Attendance + Automated Payroll - salary computation from check-in/out data (EPF/ETF, overtime, late penalties)
- Inventory Management - dual-tier stock tracking with automated reorder alerts
- ML-Powered Employee Risk Scoring Engine - the core research contribution

The system is not just an ERP. It is a decision-support artefact - the ERP collects operational data, and the ML engine transforms that data into actionable workforce risk insights.

2. Current Progress

Item	Status
Research proposal	Completed and submitted
Source of Truth document (full project spec)	Completed
Project plan with day-by-day schedule	Completed
Technology stack finalised	Completed
Research objectives & questions defined	Completed (8 objectives, 7 RQs)
Database schema design (9 collections)	Designed and documented
API endpoint architecture (~21 endpoints)	Designed and documented
System architecture and data flow	Fully designed
SLR paper search	Started - identifying 20-25 papers

3. Current Phase: Systematic Literature Review

The SLR is in active progress, targeting submission by 10 April 2026. It covers 5 thematic areas:

Theme	Papers	Covers RQs
ERP systems for SMEs (MERN/web-based)	3-4	RQ1, RQ2, RQ7
Employee attrition prediction using ML	8-10	RQ4
Social media signals in HR analytics	3-4	RQ3
Synthetic data generation (Gaussian)	2-3	RQ3
Explainable AI + Human-in-the-Loop ML	2-3	RQ5

SLR Timeline:

Date	Task
Apr 1-3	Paper search and summarisation
Apr 4-6	Draft SLR - introduction, search strategy, thematic review
Apr 7-8	Comparison table, gap analysis, research gap justification
Apr 9-10	Review, IEEE formatting, final proofread and submission

4. Research Contribution

The Research Gap

Existing systems fall into three silos:

- ERP only (SAP, Odoo) - no ML-based employee risk scoring
- ML attrition prediction only - standalone models, not integrated into ERP
- Enterprise HR analytics - expensive, not tailored for SMEs

No lightweight ERP system currently combines payroll automation, a formally documented synthetic social behaviour signal, and ML-powered employee risk scoring with a human-in-the-loop feedback mechanism.

Three Pillars of Novelty

Pillar 1: Synthetic Social Score as API Proxy

- Composite signal (0.0-1.0) using Gaussian sub-score modelling (Box-Muller transform)
- 4 sub-scores: job search (0.35), post drop (0.25), network growth (0.20), sentiment (0.20)
- Formally documented as a placeholder for LinkedIn/Facebook API data
- Schema designed for zero-migration transition to live data

Pillar 2: Integrated Risk Scoring in Lightweight ERP

- 8-feature vector from cross-module ERP data (attendance + payroll + social)
- Dual scoring: weighted composite formula (Node.js) + Random Forest classifier (Flask)
- Risk tiers: Low / Medium / High / Critical

Pillar 3: Human-in-the-Loop ML Retraining

- HR managers accept/reject ML predictions
- Accumulated decisions become ground truth for model retraining
- Bridges gap between automated prediction and responsible HR decision-making

5. Methodology - Design Science Research

Design Science Research (Peffers' DSRM) was chosen because the primary outcome is an IT artefact that solves a real-world problem.

DSRM Stage	Application in This Project
1. Problem Identification	SMEs lack integrated ERP with predictive workforce analytics
2. Objectives	Build NEXUS ERP with automated payroll, inventory, ML risk scoring
3. Design & Development	MERN stack implementation + Flask ML microservice
4. Demonstration	System walkthrough with role-based use cases
5. Evaluation	Functional testing + ML evaluation (accuracy, SHAP, confusion matrix)
6. Communication	The thesis document itself

6. Full Project Timeline

Phase	Dates	Deliverable
Phase 1: SLR + Research	Apr 1-10	SLR document ready for submission
Phase 2: Backend Dev	Apr 11-20	Full backend API with seed data
Phase 3: ML Microservice	Apr 21-25	ML pipeline - predict, train, retrain, explain
Phase 4: Frontend Dev	Apr 26-May 2	Full frontend with dashboard and risk UI
Phase 5: Evaluation	May 3-5	All evaluation metrics and charts
Phase 6: Thesis Writing	May 5-10	Complete thesis (Chapters 1-6)
Phase 7: Testing	May 11-12	Fully tested codebase + documentation
Phase 8: Delivery	May 13-16	Final handover + mentorship sessions

7. Evaluation Strategy

ML Evaluation (70% of evaluation chapter)

- Classification accuracy and confusion matrix
- Precision / Recall / F1 score per risk class
- Feature importance analysis
- SHAP explainability plots
- Before/after retraining comparison

Social Score Validation

- Gaussian distribution verification (histograms, Q-Q plots)
- Mean/std dev checks against configured parameters
- Temporal drift analysis for at-risk employees

Functional Testing

- Payroll accuracy verification

- Inventory alert response time
- API performance and security testing (JWT, OTP, RBAC)

8. Thesis Chapter Structure

Chapter	Title	Weight
1	Introduction	10%
2	Literature Review (SLR)	15%
3	Methodology (DSR)	10%
4	Design & Implementation	25%
5	Evaluation & Results	30%
6	Conclusion & Future Work	10%

Chapters 4 and 5 carry 55% of the thesis weight, with heavy focus on the ML risk scoring engine, social score model, and SHAP-based evaluation.

9. Suggested Questions for Supervisor

1. Are you satisfied with the SLR thematic structure covering 5 areas? Should I add or merge any themes?
2. For the evaluation chapter, is the planned ML metrics set (confusion matrix, SHAP, F1, retraining comparison) sufficient?
3. Do you have any concerns about using synthetic data with Gaussian modelling instead of a real dataset?
4. Should I prioritise any particular chapter or section for early draft review?
5. What formatting standard should I follow - IEEE or university-specific template?
6. Are there any additional papers or references you'd recommend for the literature review?

10. Key Points to Communicate

- The project is well-planned with a detailed day-by-day schedule covering all 8 phases.
- The research gap is clearly identified - no existing system combines all three novelty pillars.
- The methodology (DSR) is appropriate for building and evaluating an IT artefact.
- The SLR is actively in progress and on track for April 10 submission.
- All data is synthetic by design - formally modelled, statistically validated, with a documented production pathway.
- The ML component uses SHAP for explainability - aligning with responsible AI practices.
- The system includes a human-in-the-loop feedback mechanism - no automated adverse HR actions.

Prepared for supervisor meeting on 2 April 2026.